

LESSON 13.9

CREATING DISPLAYS FOR CATEGORICAL DATA



LESSON INTRODUCTION

7.MA.DP.1.5 Given a real-world numerical or categorical data set, choose and create an appropriate graphical representation.

LEARNING TARGETS

- I can create an appropriate graphical representation for a categorical data set.

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

MA.912.DP.1.1 Given a set of data, select an appropriate method to represent the data, depending on whether it is numerical or categorical data and on whether it is univariate or bivariate.

ESSENTIAL QUESTIONS

- What graphical displays are used to display categorical data?
- How can you determine whether to use bar graphs or circle graphs to display data?
- How can you interpret bar graphs and circle graphs to interpret data?

LESSON NARRATIVE

In this lesson, students consider graphical displays that best represent categorical data. Students begin by creating a bar graph and circle graph to represent a data set and then determine when it is best to use each graphical display. Students apply this knowledge in the components that follow to choose which graphical display to create to best display given data sets and justify their choices. This lesson ends with students creating either a bar or circle graph to display data and explain their choice.

COMPONENT SUMMARY

13.9.1 WARM-UP (~5-8 MIN.)

Explain why base-10 number system is useful based on given information

13.9.2 GUIDED INSTRUCTION (15-20 MIN.)

Create bar and circle graph from a data set

13.9.3 TRY IT (10-15 MIN.)

Choose a graphical representation to create and explain

13.9.4 WRAP-UP (~5-7 MIN.)

Create a graph to best display a data set and explain that choice

RESOURCES

GLOSSARY

Key Terms:

- Bar graphs
- Circle graphs

Additional Terms: N/A

MATERIALS

- Warm-Up Video
- (Optional) Calculators

ADDITIONAL PREPARATION

- 13.9.1 Warm-Up contains a video to accompany the question prompts.
- If allowing students to create graphs online, prepare websites and instructions ahead of time.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may confuse when to use a bar graph and when to use a circle graph.
- Students may still need support with creating circle graphs from Unit 7.

THINGS TO CONSIDER

- When comparing a category to the group as a whole, it is best to use a circle graph. When comparing a category to another category, sometimes it is best to use a bar chart.
- Allowing students to use calculators during the lesson will help students focus on discovery and the process of creating data displays.

LITERACY AND LANGUAGE ROUTINES

Compare and Connect

13.9.2 Guided Instruction provides an opportunity for students to engage in a compare and connect routine. Model for students how to make a connection between bar graphs and circle graphs before encouraging students to do so with a partner. This encourages meta-awareness as students identify the similarities, differences, benefits, and limitations of each display.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.2.1 Multiple Representations

Throughout this lesson, students interact with data sets displayed using various representations. Data is presented in tables, bar graphs, and circle graphs. Students utilize these displays to draw conclusions about the data, as well as conclusions about which types of graphical representations are best suited for answering specific types of questions.

DIFFERENTIATE INSTRUCTION

To reduce barriers to learning, provide alternative means for response, selection, and composition. Desmos offers an online tool to create circle graphs (or pie charts), which may be helpful for students who struggle with fine motor skills. Consider providing graph paper to assist students in creating bar graphs. Graph paper may help students keep their work organized and thus display the data in a clear and organized manner.

13.9.1 WARM-UP: VISUAL REPRESENTATIONS

QUESTIONING

Consider leveraging student experience during this section. Do you have students who speak a different language at home or have immigrated from another country? This section can engage students through their own experiences with using a standard number system across different countries and languages.



13.9.1 Warm-Up: Visual Representations

Various number systems have developed over time. In some of the original systems, symbols representing values were written one after another and then added together to determine the value represented. The Roman Numeral system added the ability to represent values with fewer symbols.



Even so, to represent 4999 in Roman Numerals is MMMCMXCIX, which is still quite long. The base-10 system used today allows for positional notation where the value of a number depends on its placement.

1. What are some reasons it is better to use the base-10 representation of the number instead of the Roman Numerals?

Answers will vary. The base-10 system allows for the value of a number to depend on its placement. The use of Roman Numerals can still make numbers quite long.

2. The symbols used for the base-10 system varied regionally for a long time. The current digits used worldwide, 0123456789, evolved from the North African region of the Arab empire. Why do you think having a standard worldwide numbering system is helpful?

Answers will vary. A standard worldwide numbering system keeps consistency from country to country. Mathematical operations will also stay the same with a standard numbering system. This ensures that an American student could go to Germany and understand the numbering system.

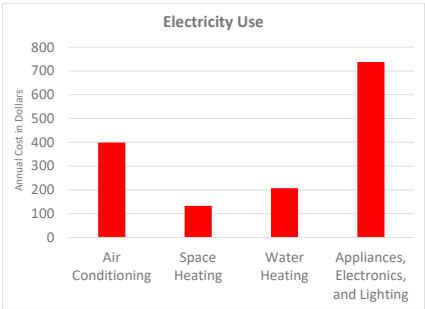


13.9.2 Guided Instruction: Graphical Representations for Categorical Data

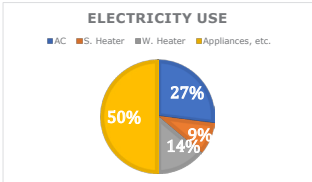
Lupe is reviewing her bills to determine if there is a way to save money by using less energy. In 2020, she paid a total of \$1477 in electricity costs to Florida Power and Light (FPL). The table shows the total cost associated with each area of electricity use in her household for the year.

Area of Electricity Use	Annual Cost, in Dollars	Percentage
Air conditioning	399	27%
Space heating	133	9%
Water heating	207	14%
Appliances, electronics, and lighting	738	50%

1. Use the data to create a bar graph for Lupe’s data set.



2. Complete the table by calculating the percentage of Lupe’s electricity costs for each area of electricity use.
3. Create a circle graph for Lupe’s data set.



4. Discuss with your partner how each data display can be useful to Lupe as she considers ways to reduce her energy bills.
Answers will vary. Half of her energy bill is from appliances, electronics, and lighting. Reducing the use of these would reduce her energy bill.
5. Complete the statements.

The bar graph displays the ☐ actual amount
☐ proportion of the total cost

Lupe pays for each area of electricity use, while the circle graph displays the ☐ actual amount
☐ proportion of the total cost

Lupe pays for each.

13.9.2 GUIDED INSTRUCTION: GRAPHICAL REPRESENTATIONS FOR CATEGORICAL DATA

DIFFERENTIATE INSTRUCTION

This component provides an opportunity to spiral review on how to calculate percentages and how to construct circle graphs from previous units in the course.

TEACHER MOVES

Compare and Connect

After students have completed the statements in question 5, consider having students share their strategies for determining whether they should use a bar or circle graph in partners or small groups. Encourage students to make comparisons between how the data is displayed in each graph and consider which information can be garnered from both and which information is specific to each graph. Students can make connections between the type of data presented and which graphical display is best suited to display that data.

13.9.2 GUIDED INSTRUCTION: GRAPHICAL REPRESENTATIONS FOR CATEGORICAL DATA (CONTINUED)

TEACHER MOVES

The purpose of this activity is to show that either graph can be useful, but sometimes one is more useful than the other depending on the purpose. The summary questions are a great way to emphasize that there are certain circumstances where one of the representations works better. It's also a good place to acknowledge that depending on what you're trying to show, the bar graph may be a better choice than the circle graph. It's not always about the data; sometimes it's about the context or purpose of displaying the data.

DIFFERENTIATE INSTRUCTION

If students have created a reference sheet to keep track of the different graphical displays and the type of data they are best suited for, encourage students to add information from the blue boxes to their reference sheets.

Lupe can use the  bar graph  circle graph to analyze whether there are any areas of electricity use that make up too large of a portion of her bill to determine where she should try to reduce her electricity consumption. She can use the  bar graph  circle graph to show her family the actual cost of each area of use as they set goals for using less energy, such as taking shorter showers or running the air conditioner at a higher temperature.

Bar graphs and circle graphs are both useful for visually representing categorical data.



Bar graphs are best when the goal is to compare categories to each other or to compare the actual values. They visually show how each part compares to other parts and display the exact counts or values for each category.



Circle graphs are best when the goal is to compare parts to a whole. They visually show each category as a proportion related to the whole.

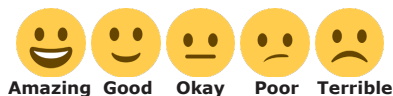
The purpose of displaying the data can also be used to determine which graph is best.



13.9.3 Try It: Graphical Representations for Categorical Data

Terrance is a server at a local restaurant and is being evaluated by his manager on his customer service skills.

After their meals, 50 customers sitting at his tables were given a survey asking to rate their satisfaction with their service on the scale shown.



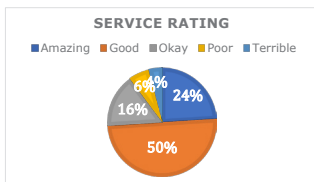
The results are shown in the frequency table.

Rating					
Frequency	12	25	8	3	2

- Complete the table to determine the percentage of customers who responded with each rating. The first row has been completed for you.

Rating	Percentage	
	$\frac{12}{50} = 0.24$	$0.24 \times 100 = 24\%$
	$\frac{25}{50} = 0.50$	$0.50 \times 100 = 50\%$
	$\frac{8}{50} = 0.16$	$0.16 \times 100 = 16\%$
	$\frac{3}{50} = 0.06$	$0.06 \times 100 = 6\%$
	$\frac{2}{50} = 0.04$	$0.04 \times 100 = 4\%$

- Use your calculated percentages to create a circle graph displaying Terrance's survey data



- With your partner, discuss some conclusions that can be drawn about Terrance's customer service based on the circle graph. Then, write three statements summarizing the conclusions discussed.

Answers will vary. Most (90%) of the customers had okay or better service. The percentage of customers that had poor or terrible service was 10%. Overall, Terrance's customers are happy with his service.

- Complete the statement.

The best graphical representation for Terrance's data is a

- ☐ bar graph because it helps us visualize how each
 ☒ circle graph

rating category compares to the whole.

- If Terrance's manager wants to compare two servers' customer service skills, explain whether it would be better to create two circle graphs or two bar graphs.
A circle graph would allow the manager to see each server's rating broken into percentages and how the ratings compare as a whole. This will be easier to make a comparison if the number of data points for each server is not the same.

13.9.3 TRY IT: GRAPHICAL REPRESENTATIONS FOR CATEGORICAL DATA

TEACHER MOVES

Consider a review of converting from a decimal/fraction to a percentage, if needed.

- How do you convert from a decimal to a percentage?
- How do you convert from a fraction to a percentage?

TEACHER MOVES

Students may struggle to determine which graph would better display data to compare two waiters' customer service skills. Ask questions to guide student thinking like:

- What information does a bar graph display? Would this be useful in comparing data from two sets?
- What information does a circle graph display? Would this be useful in comparing data from two sets?
- Is one graph better suited to compare two data sets than the other? Why or why not?
- What information does the manager want to compare? Is that more easily seen on a bar graph or circle graph? Why?

13.9.3 TRY IT: GRAPHICAL REPRESENTATIONS FOR CATEGORICAL DATA (CONTINUED)

TEACHER MOVES

Give students an opportunity to discuss a scenario where the raw numbers might be misleading to ensure they understand. For example, if 25/50 customers rate Terrance as good and 20/40 rate his coworker the same way, comparing bar graphs with the raw data might be misleading because it would seem like more customers said Terrance's service was good, when in fact the same percentage rated them this way on both surveys. It's only that the sample size was smaller for his coworker.

6. Compare your recommendation with your partner's. Discuss any differences in your reasoning.
Answers will vary.

7. Complete the statements.

If the same number of customers responded to each

server's survey, a ☐ bar graph ☐ circle graph is a better choice for

comparing these two data sets because the

☐ height of the bars ☐ size of each section in each server's graph can be easily compared side-by-side.

If a different number of customers responded to each

server's survey, a ☐ bar graph ☐ circle graph would be better because

it would compare the ☐ actual number of customers ☐ proportion of customers who

gave each response rather than the raw numbers that

could be misleading in this case.

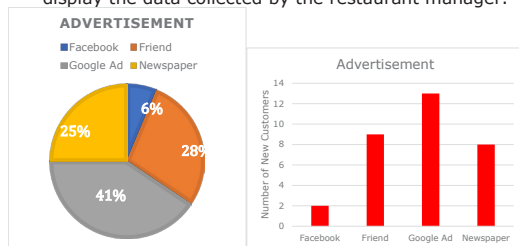


13.9.4 Wrap-Up: Ticket Out the Door

The manager at a new restaurant surveys a sample of customers to identify where their marketing efforts have been successful. The results are shown in the frequency table.

Source	Facebook Ad	Friend	Google Ad	Newspaper
New Customers	2	9	13	8

- The restaurant manager wants to make a recommendation to the owner about where to spend more money on advertisements using a data display. What graphical display is best for the manager to use to display the data?
Answers will vary. A bar graph would show results for only the sample that was surveyed. A circle graph would show the proportion of new customers compared to all those who were surveyed.
- Create the data display you chose in Question 1 to display the data collected by the restaurant manager.



13.9.4 WRAP-UP: TICKET OUT THE DOOR

DIFFERENTIATE INSTRUCTION

Use data from this formative assessment to determine which students need remediation before the next lesson.